

SpawnParamTuner — ResNet-MLP Parameter Optimization (~325K params)

INPUT: Context Embedding(32-dim) ⊗ θ(36-dim) = 68-dim

Context Embed: difficulty(4)+generator(4)+botPolicy(4)+PBBin(8)+lifecycle(8)+logPB(4) = 32

θ ∈ [0,1]³⁶: personalizations + pbTension₄ + scorings₈ + translations₅ + challenges₅ + order₂ + constructive₂ + solution₂ + specials₃

Linear(68 → 128) + LayerNorm + GELU

RESIDUAL BLOCKS — 8× ResBlock (hidden_dim=128)

RB1

FC→LN
GELU→LN
+x→GELU

RB2

FC→LN
GELU→LN
+x→GELU

RB3

FC→LN
GELU→LN
+x→GELU

RB4

FC→LN
GELU→LN
+x→GELU

RB5

FC→LN
GELU→LN
+x→GELU

RB6

FC→LN
GELU→LN
+x→GELU

RB7

FC→LN
GELU→LN
+x→GELU

RB8

FC→LN
GELU→LN
+x→GELU

Each ResBlock: Linear(128→128) → LayerNorm → GELU → Dropout(0.1) → Linear(128→128) → LayerNorm → +x → GELU
8 blocks × ~28K params each = ~224K · Pre-activation residual · Post-norm stabilization

LayerNorm(128)

OUTPUT HEADS (7 heads from shared LayerNorm output)

D(r) 20 bins

sigmoid
Difficulty curve

E(r) 20 bins

sigmoid
Delight/engage

F(r) 20 bins

sigmoid
Frustration cap

PB Break

sigmoid
PB break prob

noMove

sigmoid
No-move steps

logScore

linear
Predicted score

Survival

sigmoid
Survival prob

All heads: 128 → 64 → Output · Total head params: ~20K

INNER OPTIMIZATION

$\min_{\varphi} \mathbb{E}_{(c,\theta) \sim D} [L_{\text{total}}(f_{\varphi}(c,\theta), \text{targets})]$

Learn (c,θ) → D(r) with 15-term composite loss

α=5(shape MSE) κ=1(anchor hinge at 22 r-points) μ=1(monotonicity) δ=0.5(PB breaking) ξ=1(endpoint) τ=1(div−0.005)

OUTER OPTIMIZATION

$\theta^*_c = \operatorname{argmin}_{\theta} J(f_{\varphi}(c,\theta))$

8 LHS restarts · T=300 Adam steps (η=0.05)

Project back to [0,1]³⁶ after each step

DEPLOYMENT PIPELINE

Trained policies → policies.json · θ*_c for each context · Consumed by both SpawnPolicyRules and SpawnPolicyNet

PB curve parameters (4-dim within θ) unified across L1 (Spawn) and L2 (Params) layers